

Winding-free exact diagonalization of quantum spin ice: benchmarks and convergence criteria

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Exact diagonalization of quantum spin ice on a periodic cluster can generate noncontractible four-spin tunneling at second order, parametrically before the physical six-spin ring exchange. We formulate a nonperturbative winding-free remedy: exactly diagonalize a transported-dipole source family, canonically pull its isolated ice-descended band to a common basis, and project its zero Fourier component. At $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$, the character sequence converges from $M = 3$ to $M = 4$ within 0.50% in centered operator norm, with minimum ice overlap 0.762. The winding-free band moves the cubic-16 heat-capacity peak from $0.0255J_{zz}$ to $0.00146J_{zz}$, compared with the quantum Monte Carlo value $0.00110J_{zz}$, while retaining the exact microscopic complement. The longitudinal response simultaneously moves from $0.08065J_{zz}$ to a near-zero Γ line and $0.00535J_{zz}$ at the three allowed X points. Thus the method is nonperturbatively converged on cubic-16, but the cluster is not yet a quantitative surrogate for bulk quantum spin ice.

Motivation.—Quantum spin ice (QSI) realizes a compact $U(1)$ gauge theory in a pyrochlore magnet: the two-in-two-out constraint is Gauss’s law, defects are electric charges, and collective hexagon tunneling generates the emergent photon [1–3]. This low scale is experimentally consequential. In particular, the two heat-capacity features reported in $\text{Ce}_2\text{Hf}_2\text{O}_7$ motivate calculations that retain both the spin-ice crossover and quantum dynamics [4]. Numerically, the problem is divided by sign. Quantum Monte Carlo (QMC) reaches the zero-flux XXZ model and resolves its thermodynamics and dynamics [5, 6], whereas the opposite sign relevant to π -flux candidates has a sign problem. Exact diagonalization (ED) has no sign problem, but its periodic clusters alter the shortest allowed gauge processes.

We address that finite-size problem without replacing the microscopic model by an effective Hamiltonian. Perturbation theory is used only to identify the offending path and test its topology; production results come from exact microscopic eigenspaces. The controlled numerical limit is the transport-character resolution $M \rightarrow \infty$. On the sign-free side, winding removal must also improve convergence toward QMC. Exact removal of a known artifact is not by itself proof of convergence to the bulk.

The finite torus changes the leading process.—We study

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + S_i^- S_j^-), \quad J_{zz} > 0. \quad (1)$$

For an S^z basis state $|s\rangle$, let $Q_t(s) = \sum_{i \in t} S_i^z(s)$. The ice-rule projector is simply

$$P_{\text{ice}} = \sum_{\substack{s: Q_t(s)=0 \\ \text{for every } t}} |s\rangle\langle s|. \quad (2)$$

Thus P_{ice} is the orthogonal projector onto the subspace satisfying the ice rule on every tetrahedron. In the infinite pyrochlore lattice the shortest contractible process acting within this subspace is a hexagon. The protocol below includes both transverse couplings; the analytic scales quoted next and all present benchmark curves use the QMC-matched slice $J_{\pm\pm} = 0$. Its third-order matrix element has scale $g_6 = 12|J_{\pm}|^3/J_{zz}^2$ [1]. On a torus, however, a path may close through a periodic image. Cubic-16 contains 36 winding four-cycles in two winding classes [Figs. 1i.b) and 1i.c)] and 16 contractible hexagons [Fig. 1i.a)]. The four-spin process enters at second order with $g_4 = 4J_{\pm}^2/J_{zz}$, so $g_4/g_6 = J_{zz}/(3|J_{\pm}|)$ diverges in the nominally controlled limit. FCC-32 reduces but does not eliminate the problem: it has 24 winding four-cycles, 32 contractible hexagons, and 96 wrapping hexagons.

The distinction is visible before assigning any perturbative amplitude. Lift each exchange sequence γ to the periodic covering lattice and add its signed spin-transfer displacements. The two four-loops transport a finite dipole $\mathbf{q}_{\gamma} \neq 0$ across the torus, although they return to the same finite-cell spin configuration. By contrast, the alternating transfers on a contractible hexagon sum to $\mathbf{q}_{\gamma} = 0$. Our protocol projects the exact ice-connected band onto this winding-free ($\mathbf{q} = 0$) character, and therefore removes the former while retaining the latter.

Winding-free construction.—For an oriented bond, let $\mathbf{d}_{ij} = \mathbf{r}_j - \mathbf{r}_i - \mathbf{n}_{ij}^T L$ be its displacement and $\mathbf{c}_{ij} = \mathbf{r}_i + \mathbf{r}_j - \mathbf{n}_{ij}^T L$ its pair-flip coordinate in the periodic covering lattice. We diagonalize the explicitly sourced microscopic Hamiltonian

$$H(\boldsymbol{\theta}) = H_0 - J_{\pm} \sum_{\langle ij \rangle} (e^{2i\boldsymbol{\theta} \cdot \mathbf{d}_{ij}} S_i^+ S_j^- + e^{-2i\boldsymbol{\theta} \cdot \mathbf{d}_{ij}} S_i^- S_j^+) + J_{\pm\pm} \sum_{\langle ij \rangle} (e^{-2i\boldsymbol{\theta} \cdot \mathbf{c}_{ij}} S_i^+ S_j^+ + e^{2i\boldsymbol{\theta} \cdot \mathbf{c}_{ij}} S_i^- S_j^-). \quad (3)$$

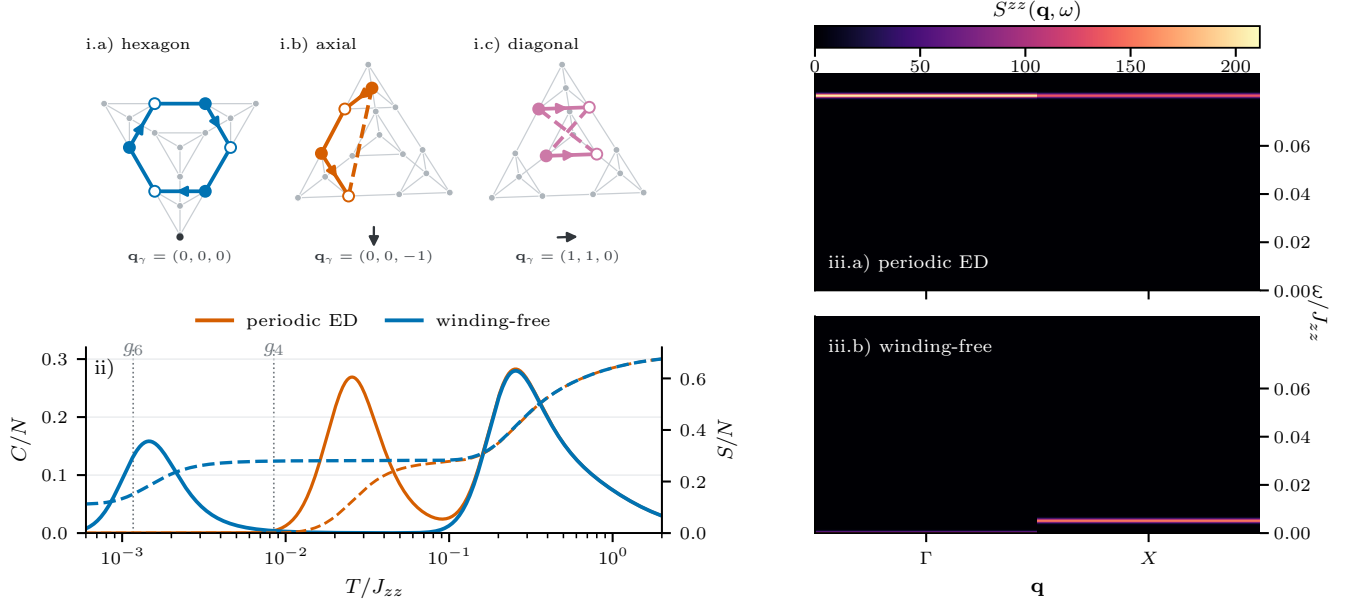


FIG. 1. Finite-torus transport and its winding-free removal on cubic-16. i.a) The contractible hexagon, viewed normal to its plane, has exact D_6 geometry and $\mathbf{q}_\gamma = 0$. i.b), i.c) Axial and diagonal winding four-loops close through periodic bonds (dashed) and carry $\mathbf{q}_\gamma \neq 0$. Gray sites and bonds show the hosting pyrochlore cluster; colored arrows are spin transfers and the dark marker or arrow is their resultant $\mathbf{q}_\gamma = 2 \sum_\ell \Delta \mathbf{p}_\ell$. ii) Specific heat (solid, left axis) and entropy (dashed, right axis) at $(J_\pm, J_{\pm\pm})/J_{zz} = (0.046, 0)$; vertical lines mark g_6 and g_4 . iii.a), iii.b) Periodic and winding-free longitudinal response on common linear scales, with X averaged over its three symmetry-equivalent representatives. The same canonically dressed microscopic S^z probe is used in both panels.

Equation (3) is the full microscopic finite-cluster Hamiltonian, not an effective Hamiltonian. No expansion or virtual-state denominator enters its definition. For a sequence γ of microscopic moves, each step ℓ acts on the stored oriented bond (i_ℓ, j_ℓ) . Its lifted polarization increment is

$$\Delta \mathbf{p}_\ell = \begin{cases} -\mathbf{d}_{ij}, & T_\ell = S_i^+ S_j^-, \\ +\mathbf{d}_{ij}, & T_\ell = S_i^- S_j^+, \\ +\mathbf{c}_{ij}, & T_\ell = S_i^+ S_j^+, \\ -\mathbf{c}_{ij}, & T_\ell = S_i^- S_j^-. \end{cases} \quad (4)$$

Multiplication of the source phases gives $e^{-i\boldsymbol{\theta} \cdot \mathbf{q}_\gamma}$ with $\mathbf{q}_\gamma = 2 \sum_{\ell \in \gamma} \Delta \mathbf{p}_\ell$. Completed ice-to-ice paths have integer $\mathbf{q}_\gamma$: contractible paths have $\mathbf{q}_\gamma = 0$, whereas the targeted paths with nonzero net winding transport have $\mathbf{q}_\gamma \neq 0$. This phase-product statement is an exact identity between microscopic matrix elements.

The symbol P_θ denotes the exact spectral projector of $H(\theta)$ onto the isolated band continuously connected to the ice-rule subspace as the transverse couplings are turned on,

$$P_\theta = \frac{1}{2\pi i} \oint_{\Gamma_\theta} (z - H(\theta))^{-1} dz, \quad \text{rank } P_\theta = \text{rank } P_{\text{ice}}. \quad (5)$$

It is distinct from the fixed ice projector. The polar map

$$W_\theta = P_\theta P_{\text{ice}} (P_{\text{ice}} P_\theta P_{\text{ice}})^{-1/2} \quad (6)$$

canonically identifies the exact band with the fixed ice-rule reference subspace. The pulled-back operator is $h(\theta) = W_\theta^\dagger H(\theta) W_\theta$. We define

$$h_M = \frac{1}{M^3} \sum_{\mathbf{m} \in \mathbb{Z}_M^3} h(2\pi \mathbf{m}/M). \quad (7)$$

Equation (7) retains $\mathbf{q} = 0 \pmod{M}$ and converges to the continuous zero-character integral as $M \rightarrow \infty$. A closing band gap or a singular $P_{\text{ice}} P_\theta P_{\text{ice}}$ is an explicit failure of the method. The construction is nonperturbative on the finite cluster because $H(\theta)$, P_θ , and the inverse square root in Eq. (6) are evaluated at the physical couplings, without a $J_\pm/J_{zz}$ or $J_{\pm\pm}/J_{zz}$ expansion. A formal expansion would recover all mixed orders in the two transverse couplings; the production calculation instead evaluates this function directly and projects its zero Fourier coefficient. Low-order Schrieffer-Wolff rows [7] only audit the loop interpretation. For thermal observables we define the hybrid replacement spectrum

$$Z_{\text{wf}} = Z_{\text{full}} - Z_{\text{periodic band}} + Z_{\text{wf band}}, \quad (8)$$

and apply the same identity to the first two energy moments. It leaves the microscopic complement intact, but it is a diagnostic spectrum rather than the spectrum of the original $H(0)$. Derivations and numerical details are given in the Supplemental Material [8].

Internal tests.—Direct row enumeration through third order gives a binary topology test. On both clusters, Eq. (7) removes 100% of winding four-loop weight and wrapping-hexagon weight and retains 100% of contractible-hexagon weight. The row classification and retained weight are defined explicitly in the Supplemental Material [8]. This is an order-two/order-three audit of the transport label; the exact-band projection itself is not truncated at either order. At $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$, cubic-16 moves from the periodic $T_p/J_{zz} = 0.02554$ to the winding-free value 0.001252. FCC-32 gives 0.008770 and 0.000608 respectively within the same third-order diagnostic. These FCC-32 values are not production thermodynamics.

The exact cubic-16 calculation uses one source convention at every M . The zero-source band has minimum ice overlap 0.762 and fixed- S^z gap $0.627J_{zz}$. All $M = 3, 4$ source points have larger overlaps and gaps, with maximum Ritz residual 3.7×10^{-12} . The centered operator changes by 6.78% from $M = 2$ to 3, but only 0.498% from $M = 3$ to 4, passing the 5% character gate. The low-order audit shows that $M = 2$ already removes the shortest winding four-loops while retaining every contractible hexagon [the retained-weight table in the Supplemental Material [8]]. It is not an all-order zero-transport projector: every nonzero $\mathbf{q} = 2\mathbf{k}$ is still aliased with zero. The sizable $M = 2$ to 3 change and subsequent $M = 3$ to 4 agreement therefore test higher-order characters, not additional removal of the audited shortest loops.

Winding-free dynamics.—We hold the physical probe fixed while changing the band Hamiltonian. Let W_0 be the exact zero-source isometry in Eq. (6) and define

$$\tilde{S}_a^z(\mathbf{q}) = W_0^\dagger \left(\sum_{j \in a} e^{-2\pi i \mathbf{q} \cdot \mathbf{r}_j} S_j^z \right) W_0. \quad (9)$$

For either $h(\mathbf{0})$ or $h_{M=4}$, we compute

$$S^{zz}(\mathbf{q}, \omega) = \frac{1}{Nd_0} \sum_{g \in \mathcal{G}} \sum_{n \notin \mathcal{G}} \sum_{a=0}^3 |\langle n | \tilde{S}_a^z(\mathbf{q}) | g \rangle|^2 \times G_\eta[\omega - (E_n - E_g)], \quad (10)$$

where $\mathcal{G}$ is the ground multiplet and $\eta = 6 \times 10^{-4} J_{zz}$. Cubic-16 has only four translation momenta, Γ and three symmetry-related X points, which reduce to the two classes shown below.

The zero-source canonical isometry reproduces the cached periodic operator to 2.7×10^{-14} relative error. The periodic ground state is unique; the winding-free ground multiplet has dimension three. The X -point centroid moves from $0.08069J_{zz}$ to $0.005372J_{zz}$, demonstrating that the construction reorganizes both energies and matrix elements rather than merely shifting a thermodynamic peak.

External QMC gate.—Huang *et al.* simulate precisely Eq. (1) at $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$ using

continuous-time worm QMC and report thermodynamics together with QMC-SAC dynamics on 8^3 primitive cells [6]. A vector extraction of their curves gives low- and high-temperature peaks at $T/J_{zz} = 0.001100$ and 0.2105 . The converged $M = 4$ result places the low peak at $0.001463J_{zz}$ and reduces the low-temperature heat-capacity RMSE from 0.0859 to 0.02327, a 72.9% improvement [Fig. 1ii]. It nevertheless misses the pre-registered 0.02 tolerance. Entropy also fails: its RMSE is 0.04509, again above 0.02 [Fig. 1ii]. The same paper provides the next dynamics gate, S^{zz} and S^{+-} at $T/J_{zz} = 0.001, 0.04$, and 0.1 ; matching its momentum points, normalization, and SAC resolution remains to be done.

Outlook.—The exact $M = 3, 4$ agreement establishes winding removal beyond low-order perturbation theory, but cubic-16 remains outside the QMC heat and entropy tolerances. The remaining campaign is a character-converged FCC-32 ABC/XYZ calculation followed by a $\text{Ce}_2\text{Hf}_2\text{O}_7$ heat-capacity fit; entropy and matched neutron channels are holdout tests, not fit normalizations.

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 - [8] (2026), see Supplemental Material for the canonical block construction, transported-dipole character projection, full-Hilbert counterterm, thermodynamic identities, geometry, and benchmark provenance.